Section 5.2: The Six Trigonometric Ratios

MATH 130 · Wednesday, November 4 · Quiz 10 on Wednesday, November 11

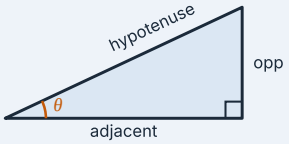
Name _____

Measured from angle θ in a right triangle:

$\sin\theta = \text{opp}/\text{hyp}$ $\quad \cos\theta = \text{adj}/\text{hyp}$ $\quad \tan\theta = \text{opp}/\text{adj}$

$\csc\theta = \text{hyp}/\text{opp}$ $\quad \sec\theta = \text{hyp}/\text{adj}$ $\quad \cot\theta = \text{adj}/\text{opp}$ (the reciprocals)

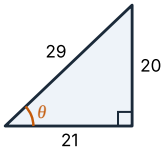
Missing side: $a^2 + b^2 = c^2$.



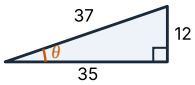
1 · All six ratios from the sides

Write each ratio as a fraction in lowest terms.

Model The legs are 20 and 21 and the hypotenuse is 29. Angle θ is opposite the side of length 20. Find all six ratios.



1 The legs are 12 and 35 and the hypotenuse is 37. Angle θ is opposite the side of length 12. Find all six ratios.



2 · Find the missing side first

Model The hypotenuse is 13 and the side adjacent to θ is 5. Find $\sin\theta$ and $\tan\theta$.

2 The hypotenuse is 25 and the side opposite θ is 7. Find $\cos\theta$ and $\tan\theta$.

3 · Variables for side lengths

Model In right triangle ABC, the right angle is at C. Side a is opposite A, b is opposite B, and c is the hypotenuse. Write $\sin A$ and $\cos A$.

3 In the same triangle, write $\cos B$, $\tan B$ and $\sec B$.

4 · Calculator values

Check that your calculator is in **degree mode** first: $\tan 45°$ should be 1. Round to four decimal places.

Model $\sin 37°$ $\quad \cos 58°$ $\quad \tan 22°$

4 $\sin 71°$ $\quad \cos 14°$ $\quad \tan 63°$

Check yourself: sine and cosine of an acute angle are always between 0 and 1. The hypotenuse is always the longest side, so csc and sec are always greater than 1. $\tan\theta$ can be any positive number.